

Continuum IDR: Hardware-Free Edge Dead-Reckoning SDK & Navigation Fallback for GNSS-Denied Environments

100% Offline Smartphone Inertial Positioning Engine with OS Mock GPS Provider Relay

TECHNOLOGY BUCKET & STACK

Kotlin

Swift (iOS CoreMotion)

Python

Edge ML (HistGradientBoosting)

Android LocationManager (Mock GPS)

6-DoF IMU Sensors

Zero-Velocity Updates (ZUPT)

Spatial Hash Road Snapping

BLE 5.0 V2V Mesh

FastAPI

React 18

0 Hardware

100% SMARTPHONE IMU

< 140 μ s

EDGE ML LATENCY

< 4.5 MB

MEMORY FOOTPRINT

28.7%

DRIFT REDUCTION (IO-VNBD)

1. EXECUTIVE DESCRIPTION

Continuum IDR is a 100% offline Edge Navigation SDK that takes over the exact millisecond GPS drops, using only the motion sensors (accelerometer & gyroscope) already inside your smartphone. No wires, no car modifications, and zero internet required.

2. HOW IT WORKS (UNDER THE HOOD)

Dual Edge ML Models (<140 μ s Latency, <4.5 MB RAM)

A lightweight **HistGradientBoosting** model predicts driving speed from a 42-moment IMU window. A companion classifier triggers **Zero-Velocity Updates (ZUPT)** at traffic lights, locking speed to 0.0 m/s so your location doesn't drift while idling.

Tuned for Indian Roads & Two-Wheeler Dynamics

Filters harsh vertical shocks from potholes and speed breakers so road ruts aren't mistaken for braking. Decouples cornering roll-lean angles $\theta = \arctan(\mathbf{v} \cdot \boldsymbol{\omega} / g)$ on two-wheelers to protect heading integrity.

Offline Road Snapping & Decentralized V2V Mesh

A localized spatial hash graph keeps your track centered on actual road vectors without cellular data, while an 18-byte peer-to-peer BLE mesh shares road hazards without internet connectivity.

★ The Killer Feature: Native Google Maps Integration

Continuum hooks directly into Android's **LocationManager.GPS_PROVIDER**. Official, unmodified Google Maps continues navigating smoothly through tunnels as if satellites were still active. When you exit the tunnel, a **2.5-second blending filter** prevents any sudden visual position jumping.

Real-World Dataset Validation

Tested and validated on the real-world **IO-VNBD** multi-driver driving dataset across **34 held-out blackout episodes**, reducing median endpoint error by **28.7%** over frozen GPS baselines.

3. ABSTRACT

Continuum IDR is a 100% hardware-free, edge-native Inertial Dead-Reckoning navigation engine and reference application. It sustains continuous, turn-by-turn vehicle tracking during total GPS outages using exclusively the built-in 6-DoF accelerometer and gyroscope already present in standard consumer smartphones. The entire platform operates strictly on-device, requiring zero external wiring, zero vehicle modifications, and zero cellular internet connectivity.

4. CORE TECHNICAL ARCHITECTURE & INNOVATION

Dual Edge ML Kinematics

Traditional double-integration of raw acceleration causes explosive cubic position drift. Continuum replaces this with a dual-stage edge ML pipeline running at 10 Hz: a **HistGradientBoosting** regressor estimates forward vehicle velocity from rolling IMU statistics, while a **LogisticRegression Zero-Velocity Update (ZUPT)** classifier instantly halts velocity integration when the vehicle stops at signals or traffic jams. Both models are compiled into a zero-dependency Kotlin tree runner executing in **<140 μ s** with **<4.5 MB RAM**.

Indian Road Surface Adaptation

Standard inertial algorithms falsely interpret vertical bumps as forward acceleration. Continuum incorporates an adaptive **a_z** vertical jerk filter to discard pothole and speed-breaker shocks. For two-wheelers, it decouples dynamic motorcycle lean angles **$\theta = \arctan(v \cdot \omega / g)$** to preserve accurate heading during aggressive turns.

Universal OS Mock GPS Relay

Synthesized coordinates are injected directly into Android's **LocationManager.GPS_PROVIDER**. Any unmodified navigation app (Google Maps, Uber, Swiggy) immediately receives real-time positions without API modifications. Upon GPS signal restoration, a **2.5-second exponential filter** smoothly blends the dead-reckoning trajectory back to satellite fixes, preventing visual snapping.

Decentralized BLE Mesh

Employs an 18-byte ad-hoc peer-to-peer BLE advertising beacon for vehicles to exchange offline hazard alerts and positioning confidence without cellular data.

5. EMPIRICAL VALIDATION & IMPACT

Benchmarked on the open **IO-VNBD** real-world vehicle dataset across **34 GPS outage scenarios** (10s to 60s durations, Driver E held-out split), Continuum IDR achieved a **28.7% reduction in median endpoint drift** compared to classical kinematic integration. By converting standard smartphones into precision navigation nodes at zero cost, Continuum delivers enterprise-grade positioning resilience for gig fleets, emergency ambulances, and daily urban commuters.